

Can Supercooled Phase Transitions Explain the Gravitational Wave Background Observed by Pulsar Timing Arrays?

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References: *Phys.Rev.Lett.* 132 (2024) 22, 221001

2024.09.12

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1. Introduction
2. The cubic potential models and supercooled phase transitions
3. Using supercooled phase transitions to explain SGWB from PTAs
 - (1) Challenges
 - (2) The state-of-the-art analysis
4. Conclusions

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1. Introduction

2. The cubic potential models and supercooled phase transitions
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(1) Challenges

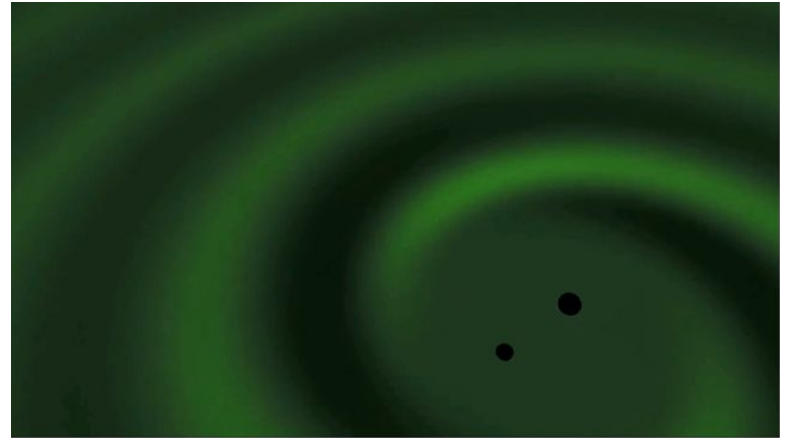
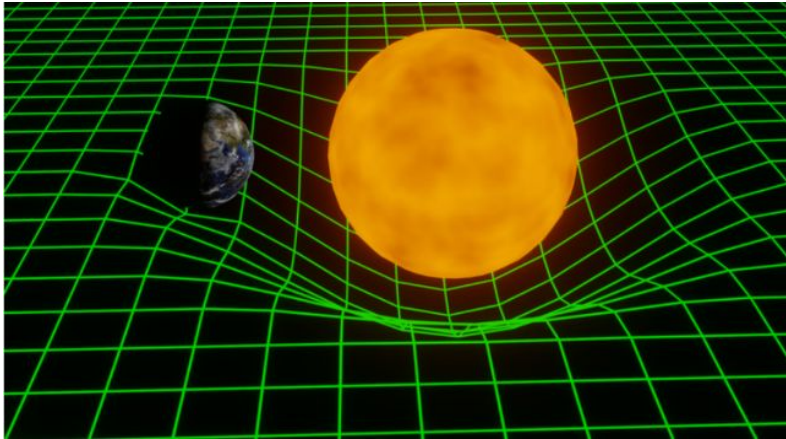
(2) The state-of-the-art analysis

4. Conclusions

Gravitational Wave ABC

1. How to generate gravitational wave (GW)?

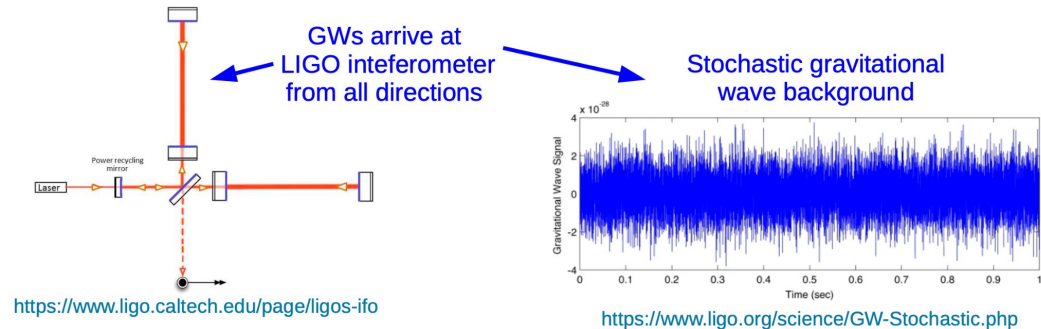
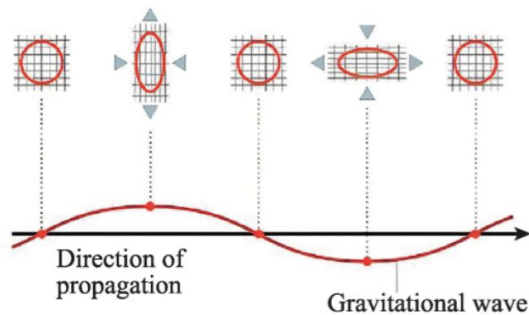
Massive objects in the universe undergoing violent motion will generate gravitational wave signals.



Gravitational Wave ABC

2. Detection principle:

The regions through which GWs pass will experience periodic stretching of space. By detecting these deformations, GW signals can be detected.

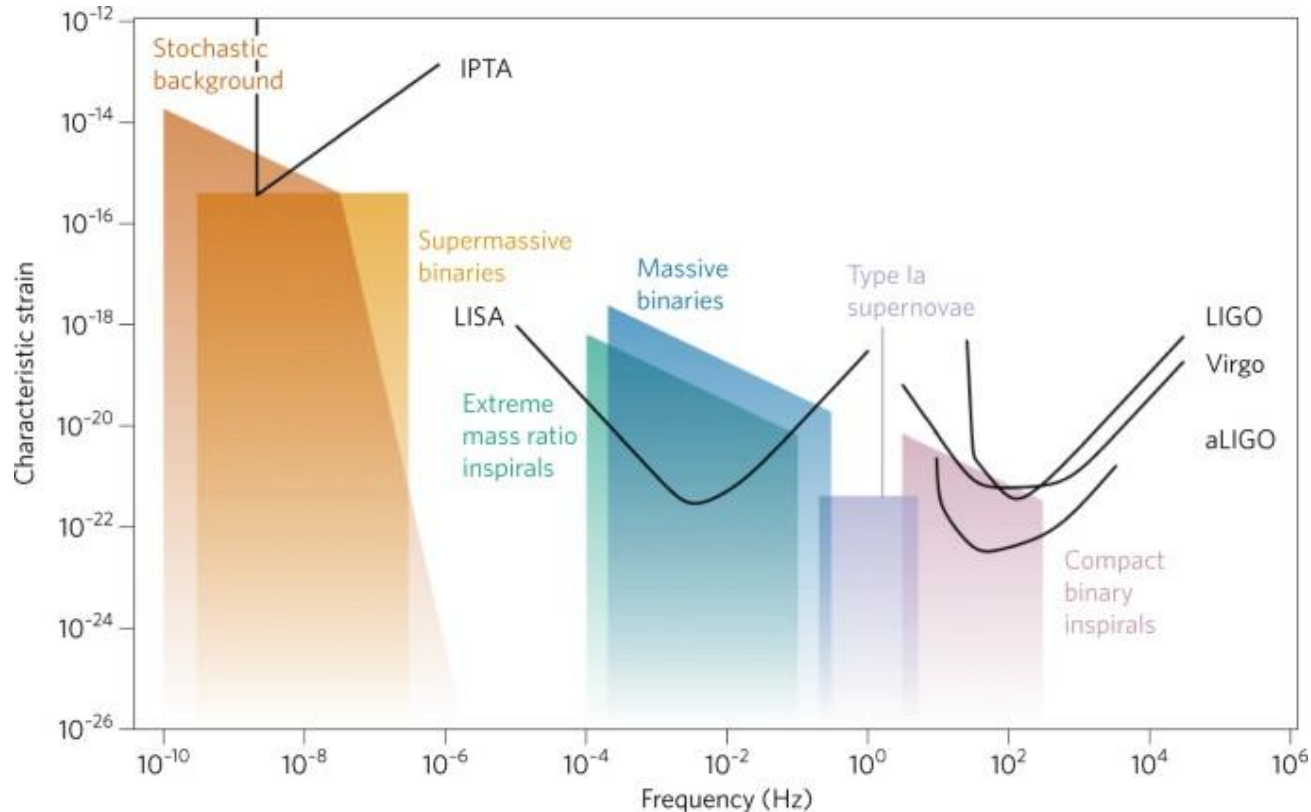


Different astrophysical processes generate GWs of different frequencies.

LIGO	$10 \sim 10^3 \text{ Hz}$	Produced by the merger of stellar-mass binary BHs and binary neutron stars	The wave source is relatively close to the Earth
LISA, Taiji, TianQin	<i>mHz</i> 10^{-3} Hz	Produced by the orbiting of binary stars and the capture of compact stars by massive BHs	Satellites form an interferometer network to conduct long-distance interferometric measurements
PTAs: NANOGrav, CPTA EPTA, PPTA	<i>nHz</i> $10^{-5} \sim 10^{-9} \text{ Hz}$		Large radio telescopes observe pulsars in the universe
BICEP2, Ali	10^{-16} Hz	Triggered by supermassive binary BHs and cosmic strings	Utilize the cosmic microwave background radiation

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Lommen, A. Pulsar timing for gravitational wave detection. *Nat Astron* 1, 809–811 (2017).



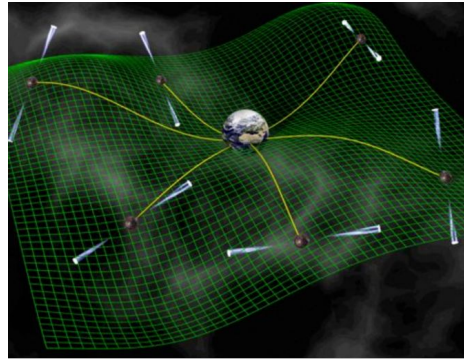
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Gravitational Wave ABC

3. Detect stochastic GWs using a pulsar timing array:

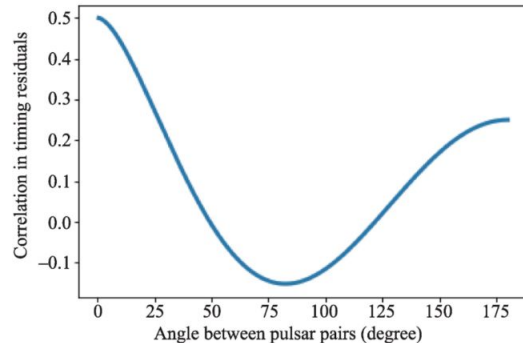
Pulsars are a type of NS with strong magnetic fields and rapid rotation. Their rotation is very stable, emitting a pulse signal at regular intervals. If unaffected by other factors, we can stably receive these signals on Earth.



Gravitational Wave ABC

3. Detect stochastic GWs using a pulsar timing array:

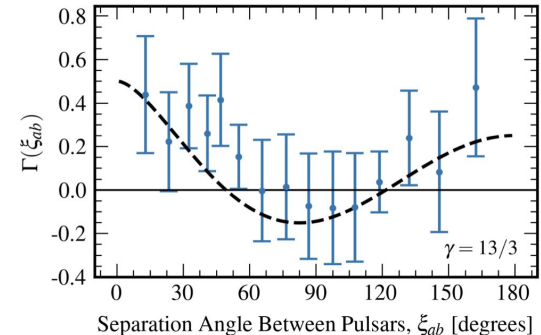
If gravitational waves pass through the region between the Earth and the pulsars, the arrival times of the pulsar signals will slightly change. Observing such regular minute changes in multiple pulsars simultaneously is an important method for detecting nanohertz stochastic GWs.



Hellings-Downs curve

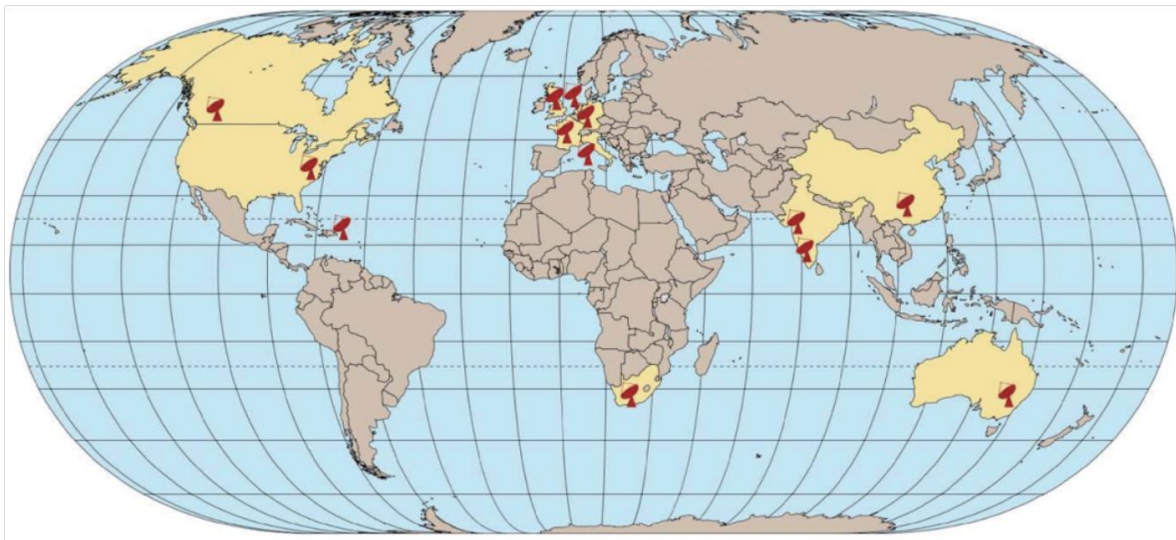
NANOGrav 15 yr Data Set

Astrophys.J.Lett. 951 (2023) 1, L8



International PTA Collaborations

EPTA monitors 18 millisecond pulsars; **NANOGrav** monitors 47 millisecond pulsars; **PPTA** monitors 27 millisecond pulsars. These three projects have also formed the **International Pulsar Timing Array (IPTA)** to jointly analyze and share data, enhancing detection sensitivity.



NANOGrav in
the USA and Canada
EPTA in Europe
PPTA in Australia
INPTA in India
SAPTA in South Africa
CPTA in China

Gravitational Wave **ABC**

3. The signal from stochastic GWs

- (1) **Definition:** When a large number of indistinguishable low frequency GWs overlap, they form a stochastic GW background signal.
- (2) **Possible origins:**
 - The overlapping GWs radiated by numerous orbiting supermassive binary BHs.
 - GW remnants from the cosmic inflation period.
 - Collisions of cosmic strings.
 - Dark sector phase transition around 100 MeV.
 - Supercooled phase transitions.

Gravitational Wave ABC

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 - **Supercooled phase transitions.**

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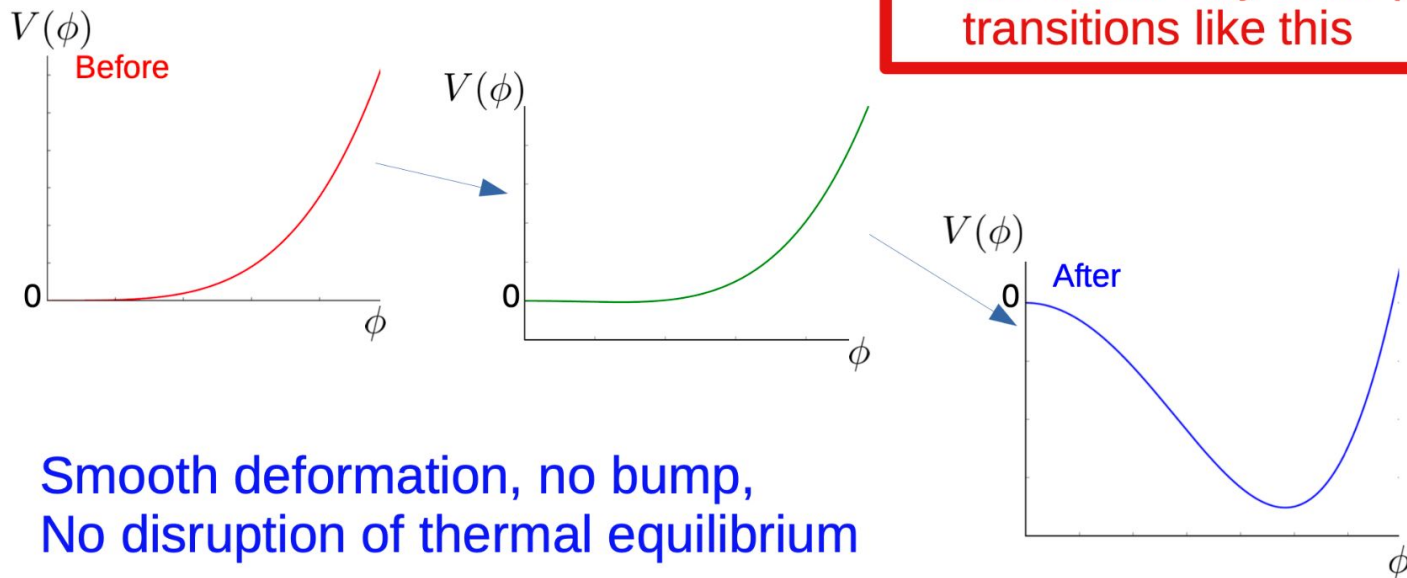
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Electroweak Phase Transitions

Finite temperature potential: $V = V(\phi, T)$

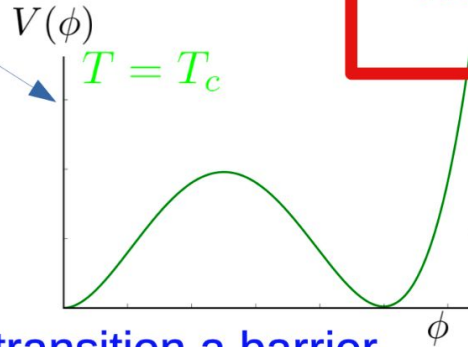
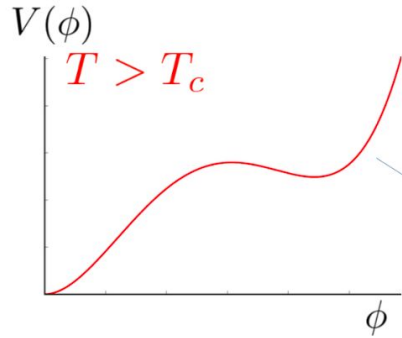
$$\text{If } V = \mu^2(T)\phi^2 + \lambda\phi^4$$

We could only have phase transitions like this

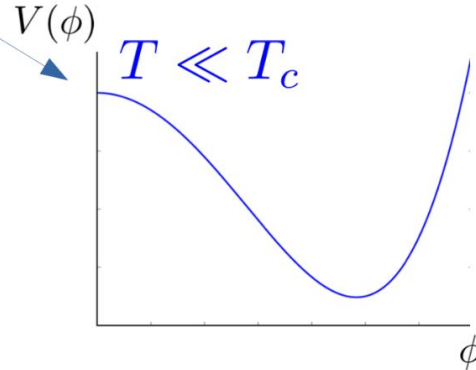


Smooth deformation, no bump,
No disruption of thermal equilibrium

First Order Phase Transitions



If $V = \mu^2(T)\phi^2 + c(T)\phi^3 + \lambda\phi^4$
We may have phase transitions
like this



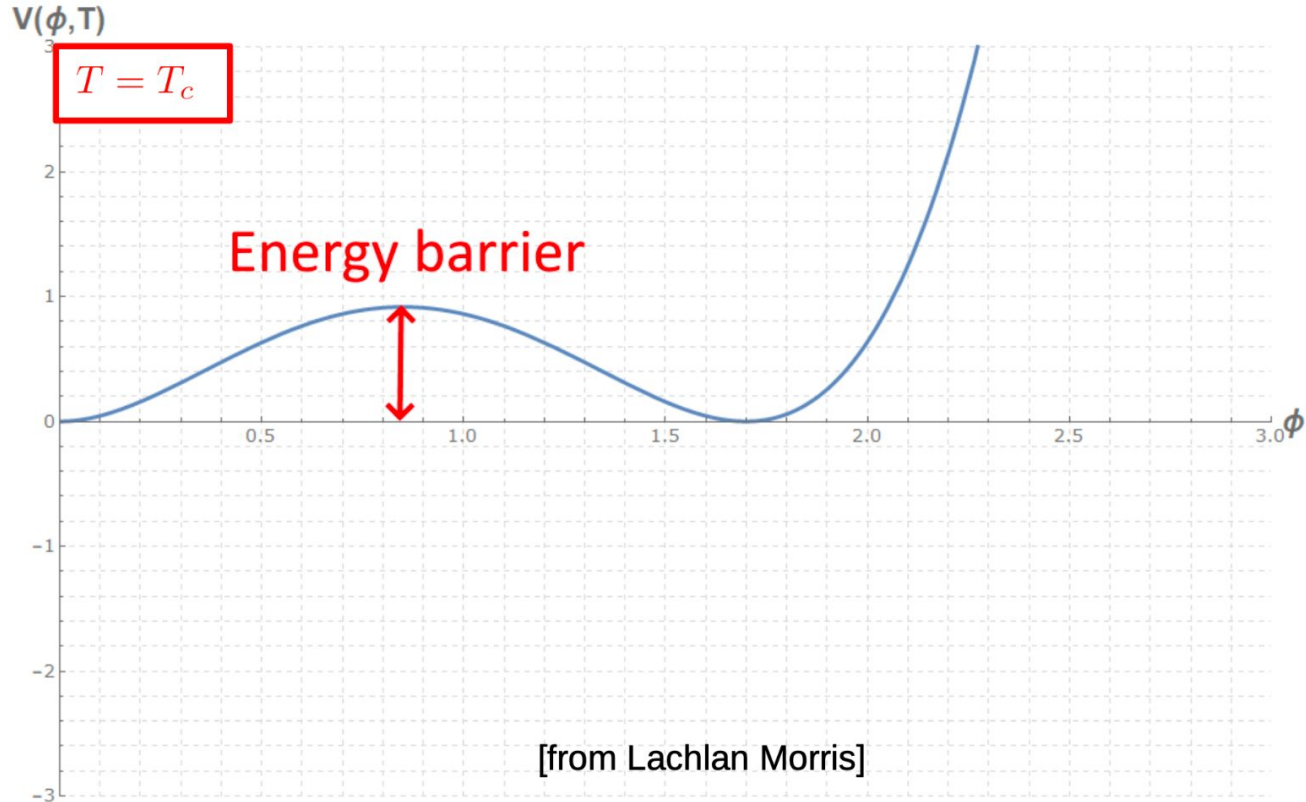
In a first order phase transition a barrier separates the two phases during the phase transition

T_c = critical temperature

$\equiv$ Temperature where V at minima are degenerate

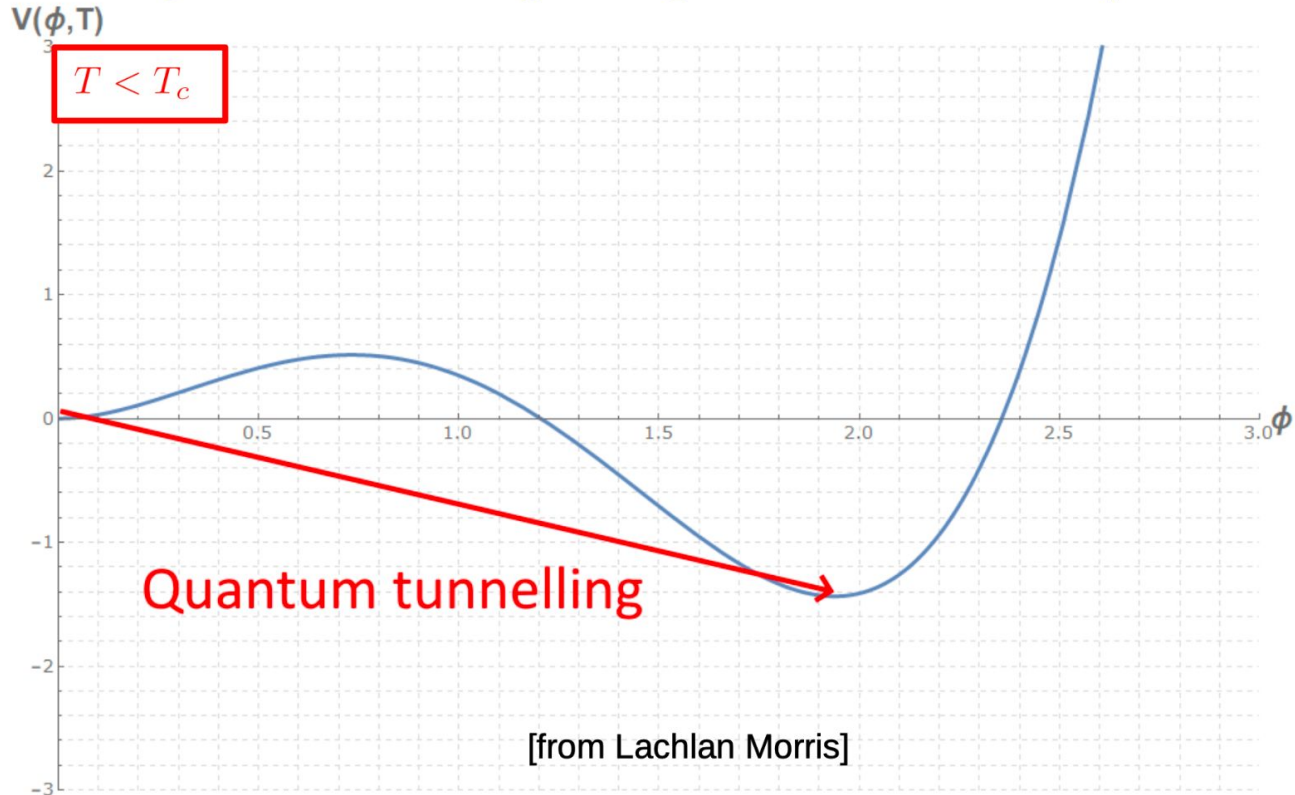
Critical Temperature

An energy barrier separates the two minima in a first order phase transition



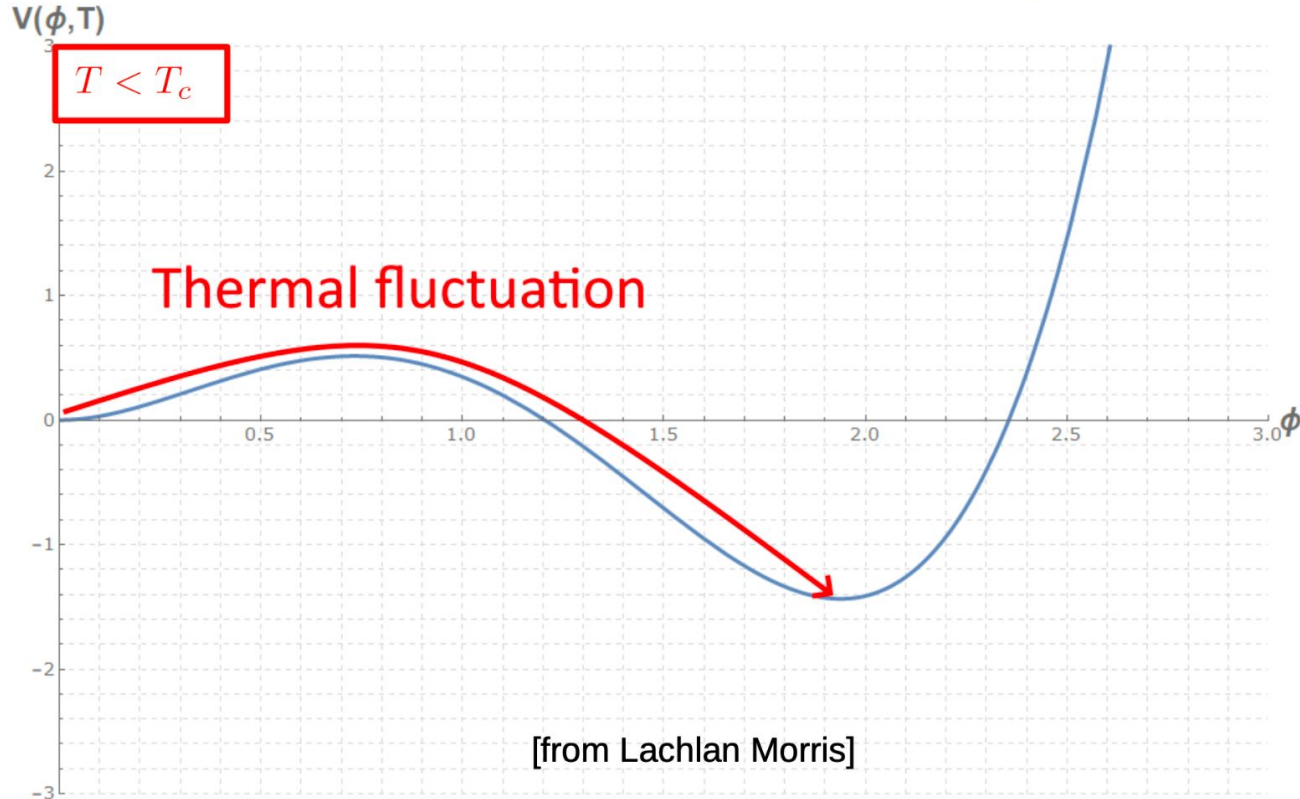
Quantum tunnelling

Quantum tunneling through the barrier is now possible



Thermal fluctuation

Thermal fluctuations over the barrier are also possible



矿泉水瓶在东北室外一摇就结冰：过冷现象

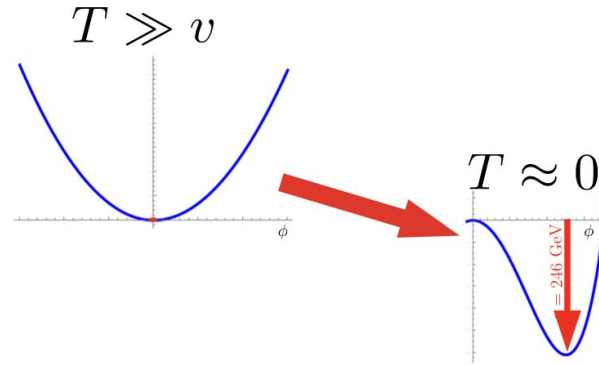


一名辽宁鞍山男子在室外拿矿泉水瓶碰撞栏杆，瓶里的水瞬间结冰，引网友热议。这实际上是，温度低于凝固点但未凝固的液体不稳定，通过摇晃碰撞迅速凝固。

材料来源：澎湃新闻

Idea:

Take the EW phase transition



$$T_{EW} \sim \mathcal{O}(100\text{GeV}) \longleftrightarrow f^{\text{peak}} \sim 10^{-5} \text{ Hz}$$

But **supercool** down to

$$\mathcal{O}(100\text{MeV}) \longleftrightarrow f^{\text{peak}} \sim 10^{-9} \text{ Hz}$$

Archetypical example: A. Kobakhidze, C. Lagger, A. Manning and J. Yue, EPJ.C 77 (2017) 570 [1703.06552] cited by NANOGRAV.

Searching for Gravitational Waves from Cosmological Phase Transitions with the NANOGrav 12.5-Year Dataset

NANOGrav Collaboration • Zaven Arzoumanian (CRESST, Greenbelt and NASA, Goddard) et al.

Phys.Rev.Lett. 127 (2021) 25, 25 • e-Print: [2104.13930](#) • DOI: [10.1103/PhysRevLett.127.251302](#)

fusion background. However, other more speculative GW sources in the PTA frequency range include cosmic strings [14, 15], a primordial GWB produced by quantum fluctuations of the gravitational field in the early universe, amplified by inflation [16–18], and cosmological phase transitions [19–23], the latter of which is the subject this study.

The NANOGrav 15 yr Data Set: Search for Signals from New Physics

NANOGrav Collaboration • Adeela Afzal (Munster U. and Quaid-i-Azam U.) et al.

Astrophys.J.Lett. 951 (2023) 1, L11 • e-Print: [2306.16219](#) • DOI: [10.3847/2041-8213/acdc91](#)

Instead, we conclude that the reconstructed posterior distribution of T_* is compatible with phase transition scenarios that have been discussed in the literature as a possible source of GWs in the PTA band: (i) BSM models in which the chiral-symmetry-breaking phase transition in quantum chromodynamics (QCD) is a strong first-order phase transition (see, e.g., Li et al. [2021](#); Neronov et al. [2021](#)), and (ii) strong first-order phase transitions in a dark sector composed of new BSM degrees of freedom (see, e.g., Nakai et al. [2021](#); Ratzinger & Schwaller [2021](#)). In view of the NG15 data, both of these options for the particle physics origin of the phase transition signal remain viable. A third option may consist in a strongly supercooled first-order electroweak phase transition (Kobakhidze et al. [2017](#)).

Bubble nucleation

Bubbles of the new phase
form at random locations



[image: from Lachlan Morris]

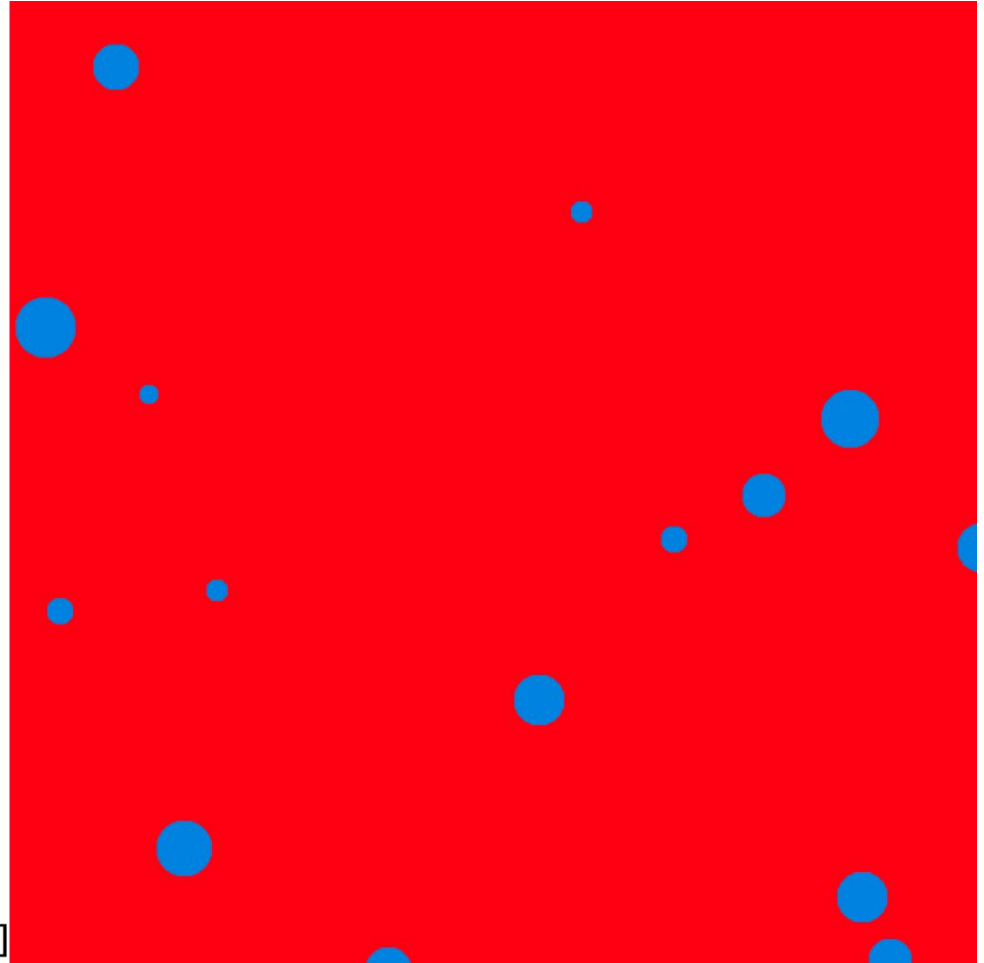
Bubble nucleation

Bubbles of the new phase
form at random locations

The bubbles that already formed
grow in size

while more bubbles nucleate

[image: from Lachlan Morris]



Bubble nucleation

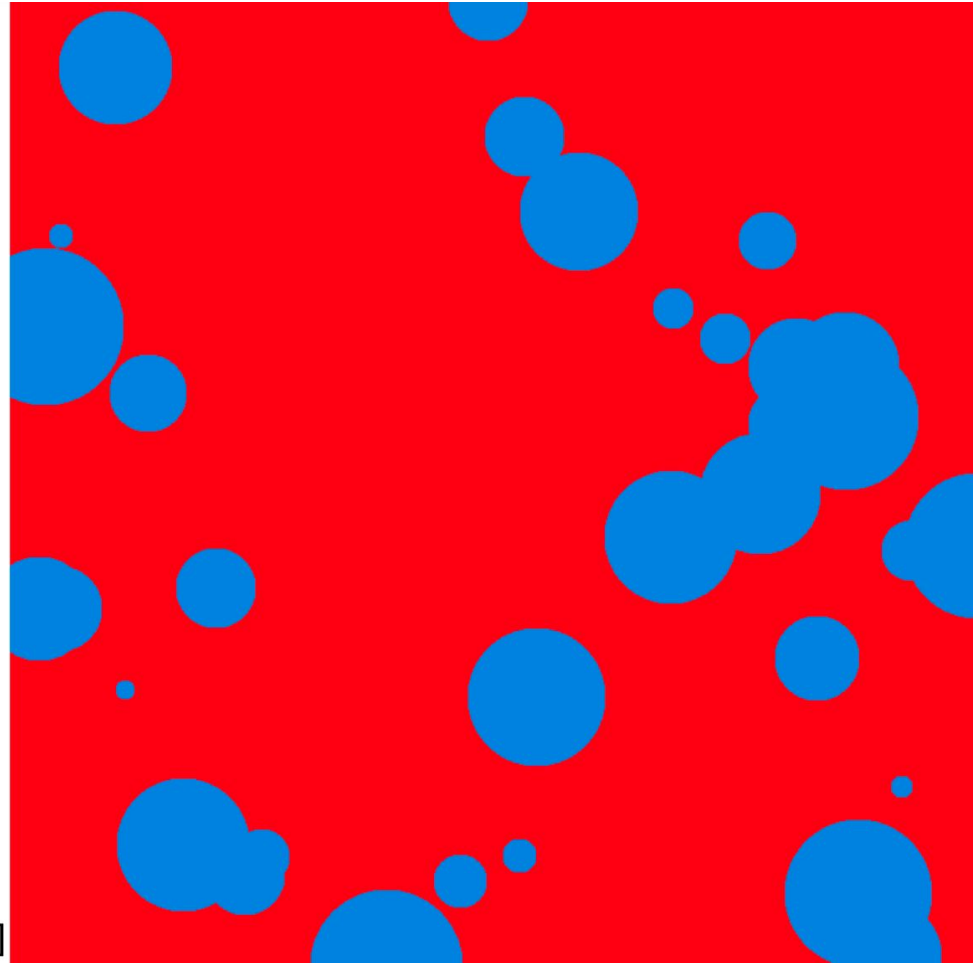
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As the bubbles grow, and the number increases, collisions become more likely

[image: from Lachlan Morris]



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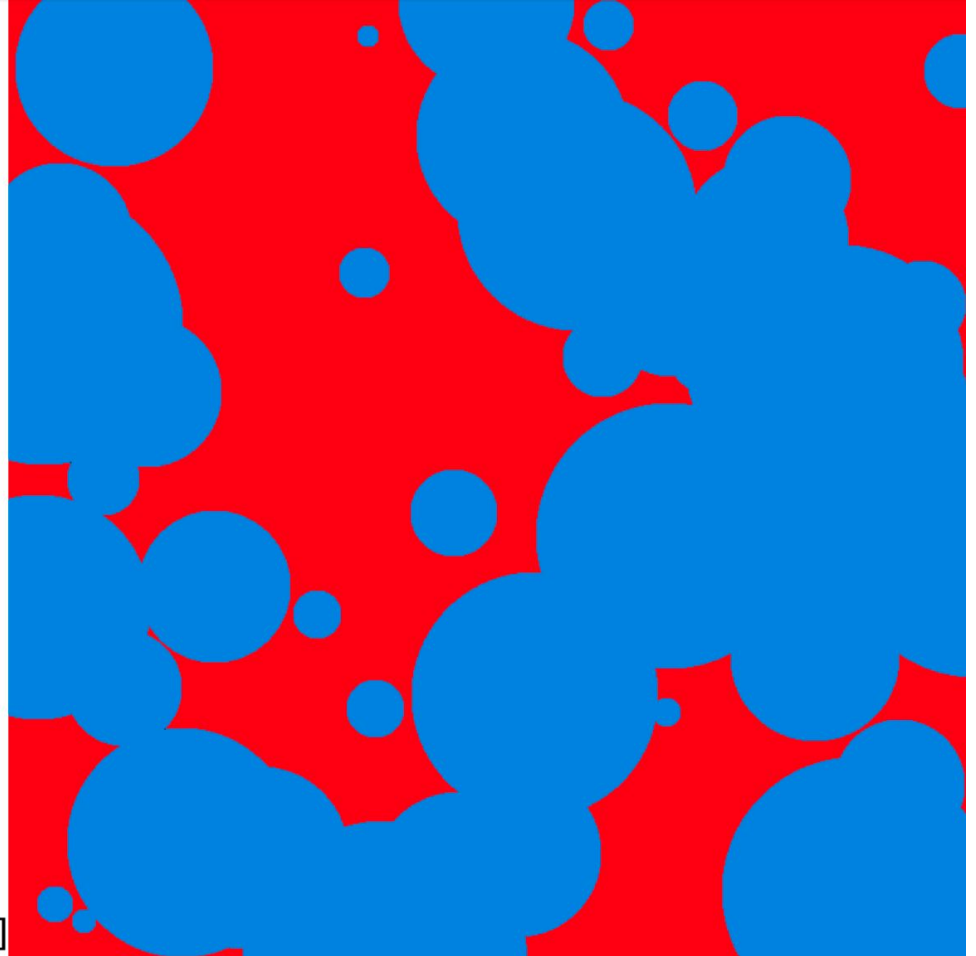
The bubbles that already formed grow in size

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As the bubbles grow, and the number increases, collisions become more likely

And more and more of the space is converted to the true vacuum

[image: from Lachlan Morris]



Bubble nucleation

Bubbles of the new phase
form at random locations

The bubbles that already formed
grow in size

while more bubbles nucleate

As the bubbles grow,
and the number increases,
collisions become more likely

And more and more of the space is
converted to the true vacuum

Until almost all the space is in the
true vacuum



Nucleation temperature

- According to the above pictures, if the decay probability is high enough, bubbles of true vacuum nucleate and expand in the surrounding symmetric phase.
- The nucleation temperature is defined as the temperature at which most of the bubbles are produced.

Nucl.Phys.B 216 (1983) 421

- Quantitatively, the nucleation temperature is given by $N(T_n) = 1$

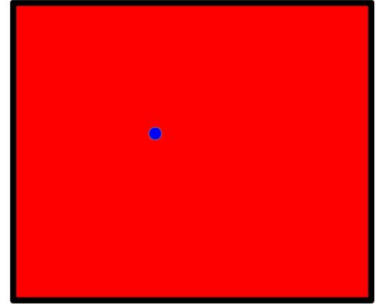
$$N(T) = \int_T^{T_c} dT' \frac{\Gamma(T')}{T' H^4(T')} \begin{array}{l} \xrightarrow{\text{Bubble nucleation rate}} \\ \xrightarrow{\text{Hubble parameter}} \end{array}$$

- The nucleation temperature is frequently used for evaluating GW signals but it may not exist ...
and for slow transitions it decouples from the other temperatures

Does the Phase transition complete?

Many studies only check **nucleation**

Nucleation: one bubble per Hubble volume

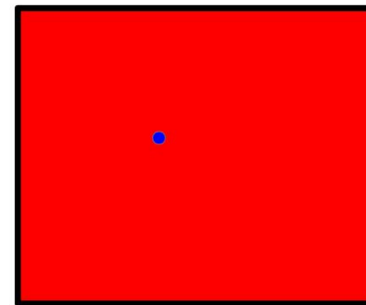
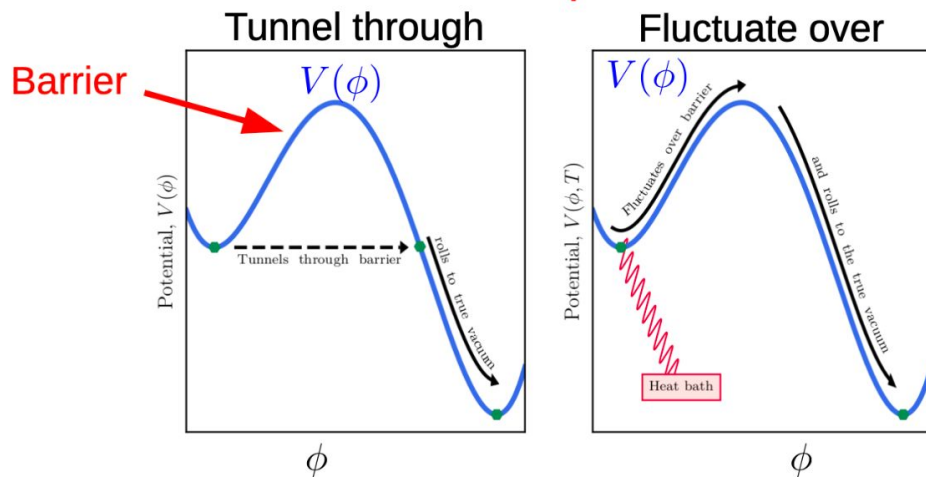


Hubble volume

Does the Phase transition complete?

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Nucleation: one bubble per Hubble volume



Hubble volume

If the barrier dissolves quickly with temperature

→ Exponential nucleation rate → Bubbles rapidly fill space

"Fast transition" or **"low supercooling"**

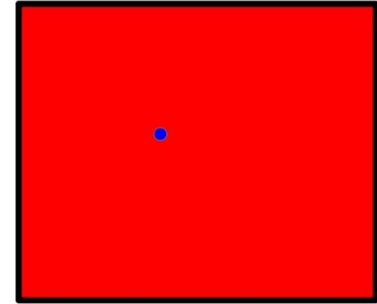
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Nucleation: one bubble per Hubble volume

Not sufficient for scenarios with a lot of **supercooling**,

If the barrier persists to low temperatures,
→ nucleation rate can reach a maximum



Hubble volume

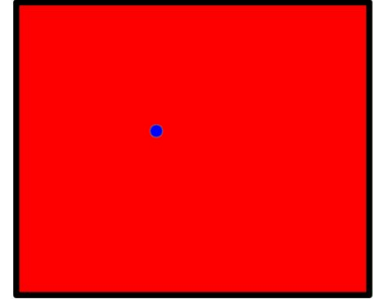
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Hubble volume

For such slow transitions we need the **false vacuum fraction** $P_f \rightarrow 0$

$$P_f(T) = \exp \left[-\frac{4\pi}{3} \int_T^{T_c} \frac{dT'}{T'^4} \frac{\Gamma(T')}{H(T')} \left(\int_T^{T'} dT'' \frac{v_w(T'')}{H(T'')} \right)^3 \right]$$

Stochastic so actually check:
 $P_f < \epsilon$

TransitionSolver JCAP 03 (2023) 006

bubble wall velocity

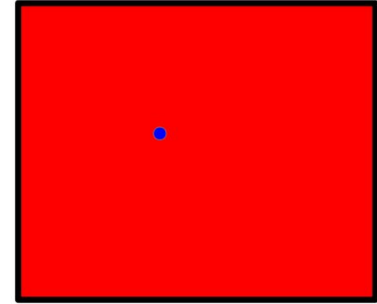
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Warning: even this is not enough because space is expanding

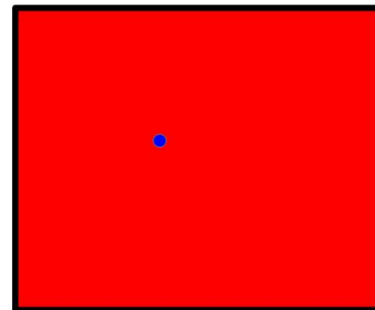
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Stochastic so actually check:
 $P_f < \epsilon$

Account for expansion of space-time and check $\frac{d\mathcal{V}_f^{\text{phys}}}{dT} < 0$ JCAP 03 (2023) 006

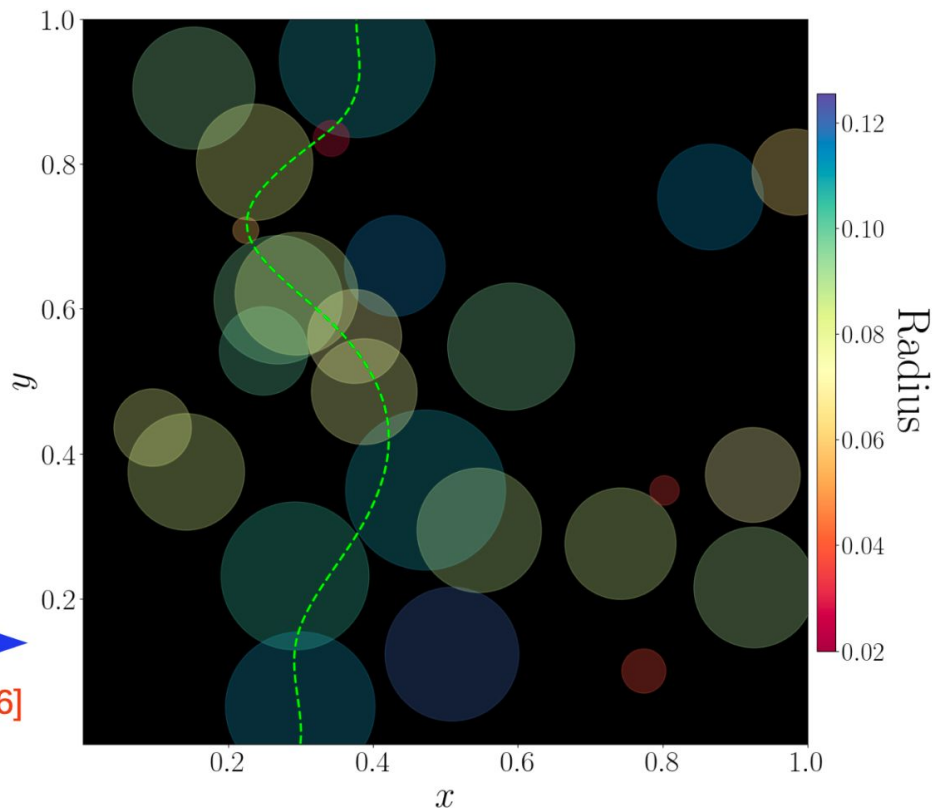
Percolation tempearture

$$T_p: P_f(T_p) = 0.71$$

- Percolation is when there is a connected path between bubbles across the space
- Strongly linked to bubble collisions
- Good choice for a temperature at which to evaluate the GWs spectrum

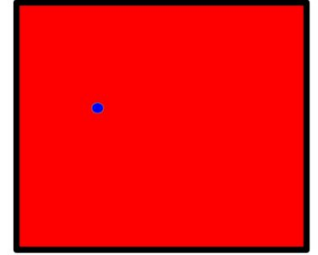
Example from simple simulation →

[PA, C. Balázs, L. Morris, JCAP 03 (2023), 006]



Temperature dependence

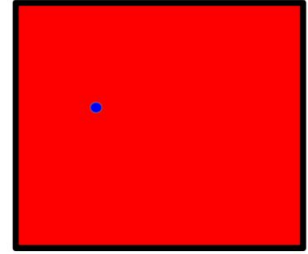
Nucleation temperature is a **bad** temperature to use
- not connected to bubble collisions



Temperature dependence

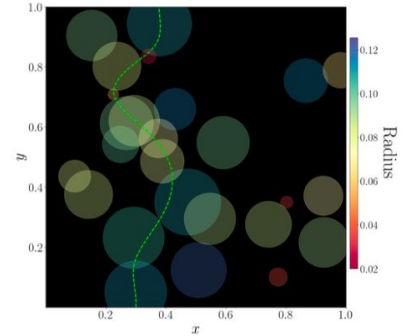
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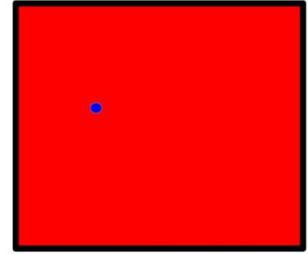
Percolation is directly defined in terms of contact between bubbles

Percolation temperature is **much better**, but...



Temperature dependence

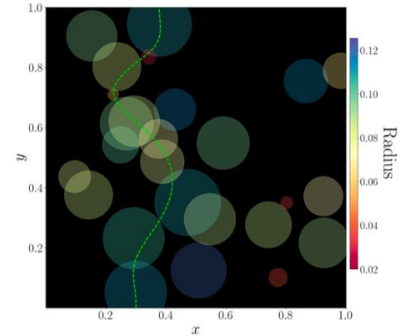
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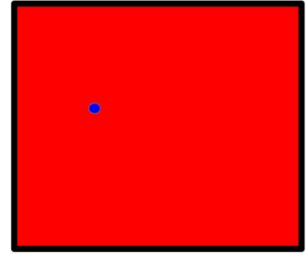
Percolation temperature is **much better**, but...

We still don't know exactly correct temperature and...



Temperature dependence

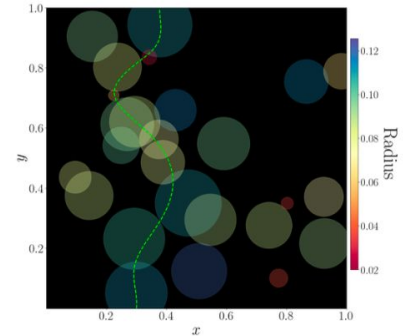
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Percolation is directly defined in terms of contact between bubbles

Percolation temperature is **much better**, but...

We still don't know exactly correct temperature and...



Percolation criteria $P_f(T_p) = 0.71$ does not account for expanding space time

→ Temperature dependence represents a significant uncertainty

False vacuum fraction $\longrightarrow$ several important milestone temperatures

Completion temperature: $T_f: P_f(T_f) = 0.01$

Percolation temperature: $T_p: P_f(T_p) = 0.71$

The instant when a significant volume of the Universe has been converted from the symmetric to the broken phase.

The nucleation temperature is instead given by $N(T_n) = 1$

$$N(T) = \int_T^{T_c} dT' \frac{\Gamma(T')}{T' H^4(T')}$$

The nucleation temperature is frequently used for evaluating GW signals but it may not exist...

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Challenges

- First, the phase transition does not complete for the low temperatures associated with a nHz signal.
- Second, the energy released by the phase transition reheats the Universe to about the new physics scale and this can rule out attempts to solve the completion problem.
- Two key criteria for the challenges of fitting a nHz signal with this cubic potential:
 1. **Realistic percolation:** Having a percolation temperature and that the physical volume of the false vacuum is decreasing at the onset of percolation;
 2. **Having a completion temperature:** A temperature at which the false vacuum fraction falls to below 1%.

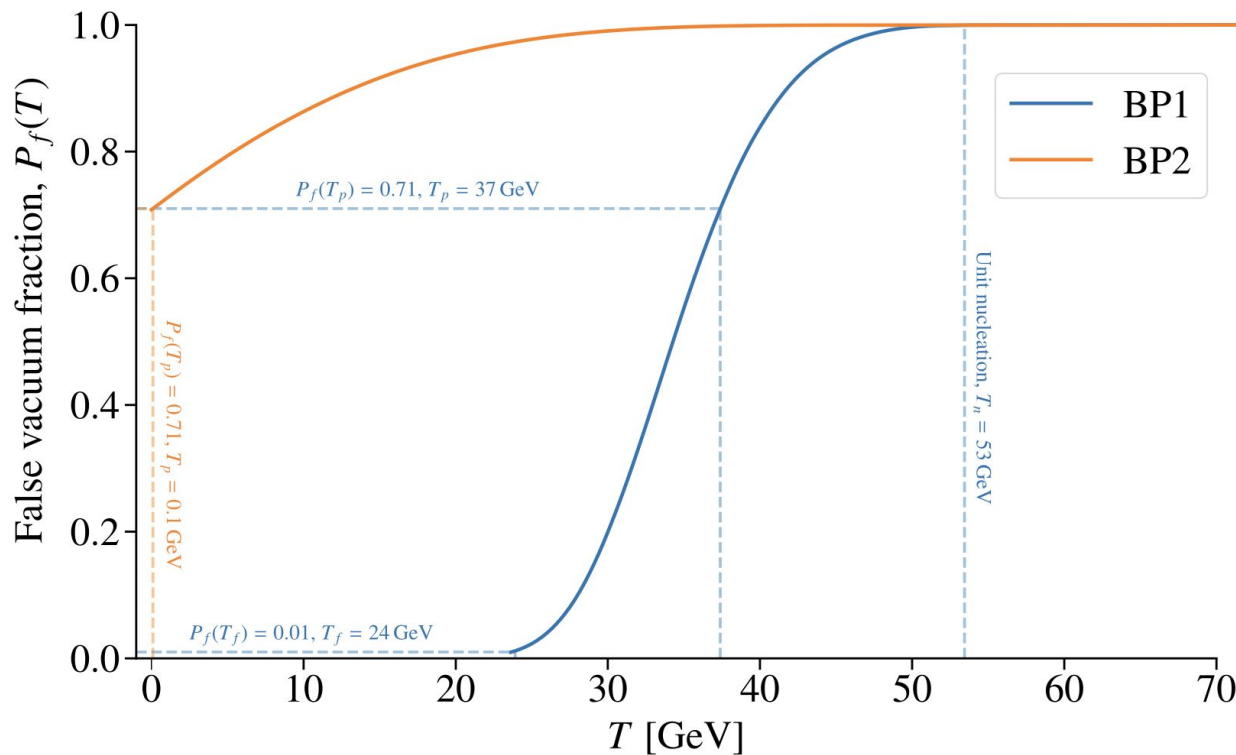
$$\text{BP1} : \kappa = -117.96 \text{ GeV}$$

- BP1 resulted in the most supercooling for which the transition satisfies both criteria, though it fails to supercool to sub-GeV temperatures;
- The physical volume of the false vacuum starts decreasing at exactly the percolation temperature.

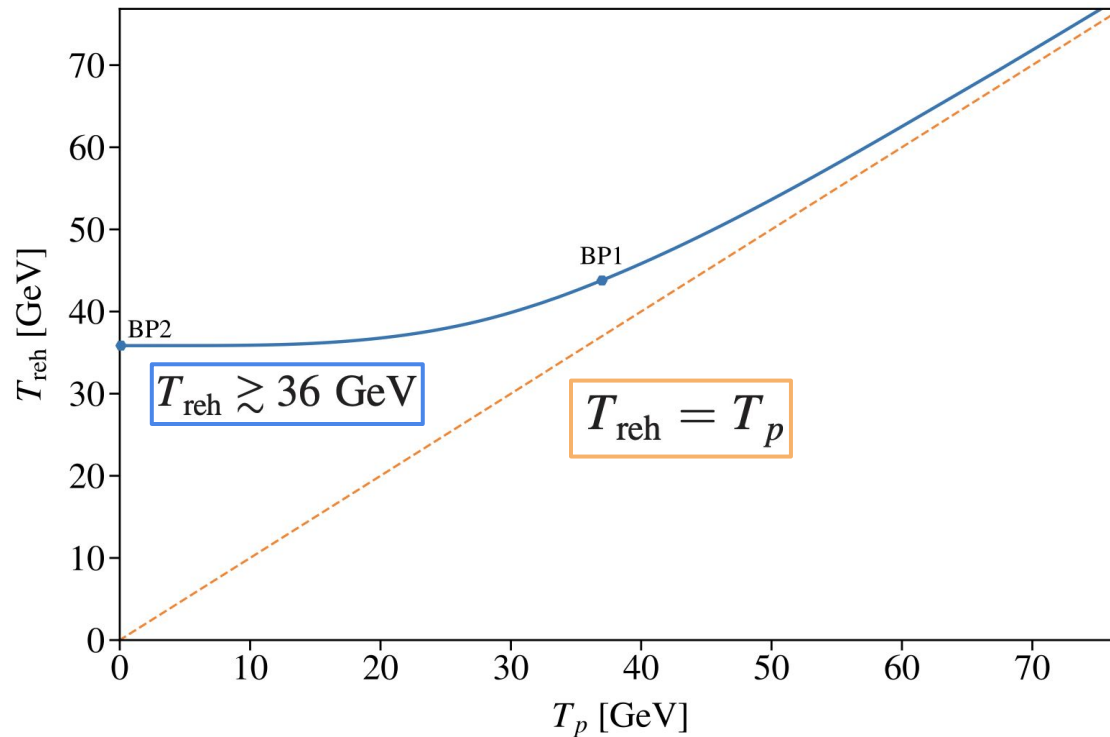
$$\text{BP2: } \kappa = -118.67 \text{ GeV}$$

- BP2 results in stronger supercooling with a nominal percolation temperature of 100 MeV but no completion temperature.
- Although BP2 was chosen so that percolation was estimated to begin at 100 MeV, it violates our first criteria and the space between bubbles continues to expand below 100 MeV. Thus, despite a nominal percolation temperature, percolation could be unrealistic.
- Without significant percolation of bubbles, the phase transition would not generate a SGWB.

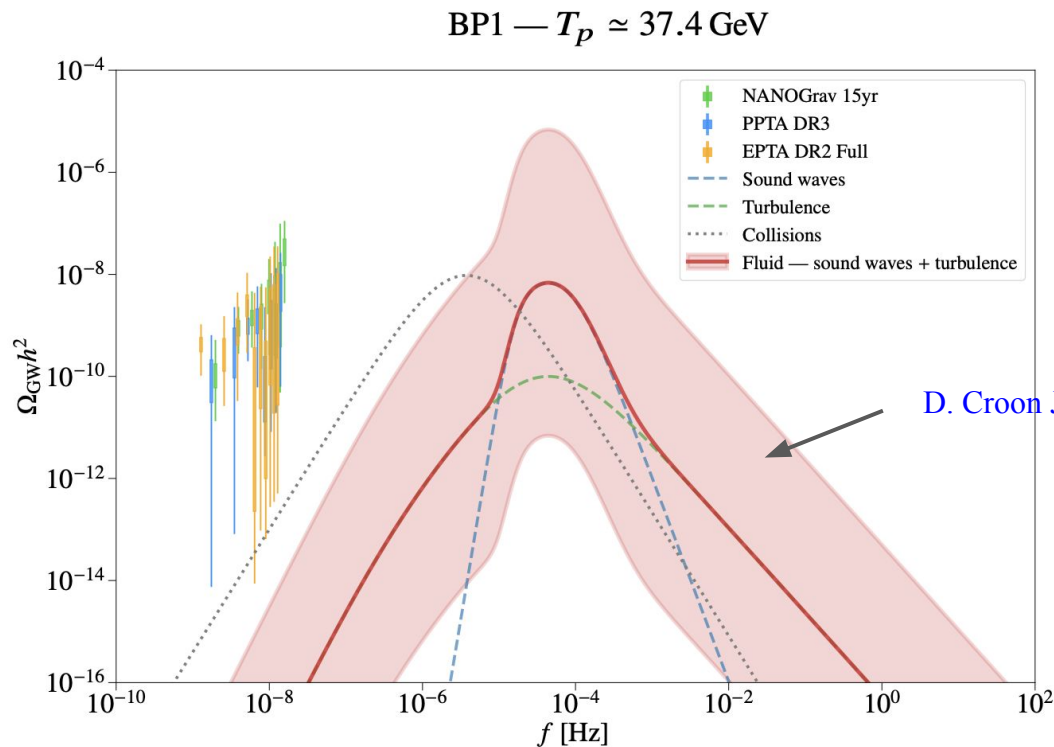
Challenge 1: Percolation and Completion



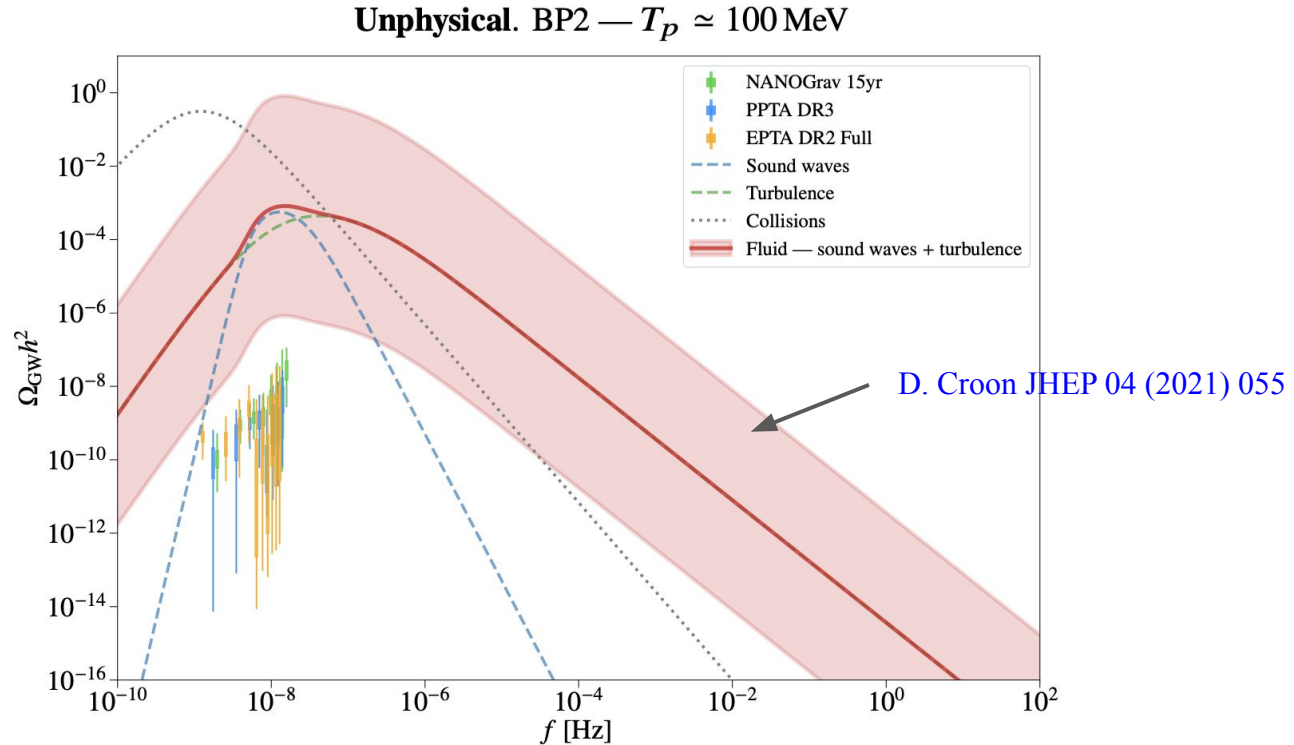
Challenge 2: Reheating



The SGWB from BP1: strongest supercooling for which the FOPT completed



The SGWB from BP2: strongest supercooling for which the FOPT has a percolation temperature though it doesn't compete and percolation is questionable



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Conclusions

1. Supercooled FOPTs are an interesting explanation of the nHz SGWB recently observed by several PTAs, as they could connect a nHz signal to the electroweak scale.
2. However, there are two major difficulties that can affect supercooled explanations:
 - Percolation and Completion
 - Reheating

Conclusions

3. Although the peak frequency could be reduced to nHz, supercooled phase transition don't complete. In contrast, completing the phase transition would NOT lead to a nHz signal.
4. These issues are quite generic and they should be carefully checked in supercooled explanations.

An interview for this topic from Phys.org

Webpage: <https://phys.org/news/2024-06-supercooled-phase-transitions-gravitational.html#>

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FEATURE



Editors' notes

Supercooled phase transitions: Could they explain gravitational wave signals?

by Tejasri Gururaj , Phys.org

A new [study](#) published in *Physical Review Letters* explores the possibility that a strongly supercooled, first-order phase transition in the early universe could explain gravitational wave signals observed by pulsar timing arrays (PTAs).

Thank you
for your attention

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